

Before the
U.S. COPYRIGHT OFFICE, LIBRARY OF CONGRESS

In the Matter of
Software-Enabled Consumer Products Study
Docket No. 2015-06

Comments of Electronic Frontier Foundation

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I. The Commenting Party

The Electronic Frontier Foundation (EFF) is a member-supported, nonprofit, public interest organization dedicated to ensuring that copyright law advances the progress of science and the arts and enhances freedom of expression. Founded in 1990, EFF represents tens of thousands of dues-paying members, including consumers, hobbyists, computer programmers, entrepreneurs, students, teachers, and researchers, who are united in their desire for a balanced copyright system that provides adequate incentives for creators, facilitates innovation, and ensures broad access to information in the digital age.

The following comments primarily relate to the Copyright Office's fourth inquiry: "Whether, and to what extent, legitimate interests or business models for copyright owners and users could be undermined or improved by changes to the copyright law in this area," but also have relevance to the overlapping topics described in the second and fifth inquiries (namely, the frustration of legitimate uses and intersections with other areas of law).

II. EULAs and Terms of Service Threaten the Balance Sought by Copyright Law

Traditionally, once a person has purchased a product, she has been free to use it however she sees fit. Purchasers have also been free to choose competitive add-on or alternative technologies that interoperate with the goods they buy, because innovators were able to develop and distribute such technologies.

Limitations on copyright have proven essential to ensuring these expectations can still be met where, as it increasingly the case, the product contains embedded software. Properly interpreted, copyright law seeks to balance the rights granted to incentivize authors of original works on the one hand, and the rights of subsequent authors and innovators, as well as market competition, access to knowledge, and the freedom of speech on the other. In particular, fair use, the idea/expression dichotomy, first sale, and Section 117 rights have ensured that device owners can learn from, tinker with, and lend or sell their property, and that competitors can enter the market with interoperable products.

As a practical matter, however, manufacturers of software-enabled devices often seek to upend that balance and impose a series of restrictions on the use of the software embedded in the device. Those restrictions may be technical, such as digital rights management technology. EFF will address such restrictions in its comments in response to the Copyright Office's contemporaneous study on Section 1201 of the Digital Millennium Copyright Act.

Equally pernicious, however, are the non-negotiated contracts of adhesion attached to the software embedded in user devices, which commonly take the form of End User License Agreements (EULAs) or Terms of Service (ToS). When those contracts include terms that strip users of fundamental speech rights or other rights enacted to protect competition and innovation, enforcing those terms as if they were freely-negotiated would undermine the balance sought by copyright law.

When rightsholders first sought to impose restrictions on purchasers by prohibiting the resale of books for less than a dollar, the doctrine of first sale emerged to prevent vendors from unilaterally enlarging the copyright monopoly. Now, however, rightsholders commonly use non-negotiated "contracts" to prevent the resale or lending of copyrighted works and software-enabled devices. Using the flexibility of such contracts, rightsholders have also written terms to prevent the use of third-party accessories, repair, reverse engineering, preparation of fair use derivative works, and enjoyment of other rights protected under copyright and other law.

Contract law provides bases to rein in some abusive practices, but relying on state law has led to inconsistent results. Furthermore, most such contracts of adhesion now mandate arbitration, which further limits the opportunity for courts to develop the law. Federal protections are needed in light of expanded attempts to erase statutory and constitutional rights via one-sided, non-negotiated "agreements."

A. EULAs and Terms of Service Seek to Erase Important Rights of Device Owners

Copyright holders have repeatedly attempted to use one-sided terms to extend their control beyond the rights granted to them by the law. For example, at the turn of the 20th century, book publishers tried to impose a minimum resale price on books by putting a notice in every copy, an effort that was rebuffed by the establishment of the first sale doctrine. Since then, Congress and the courts have consistently rejected efforts to undermine copyright's first sale principle, but have not yet adequately addressed EULAs.

Congress has taken steps to address specific issues. For example, the Consumer Review Freedom Act,¹ would forbid and render unenforceable contract provisions that restrain customers' freedom to post negative reviews.² The FTC has also filed a complaint against a company called Roca Labs for attempting to use such "gag clauses" in its customer contracts.³

However, the problems go far beyond those specific issues. Vendors seek to use Terms of Service and EULAs to strip device owners of rights that protect not only individual interests in expression and autonomy, but also societal interests in innovation and competition.

Terms of Service and EULAs employ two basic strategies in their attempts to re-write the law. The first strategy involves a purported waiver of the rights of the device owner to engage in noninfringing uses, such as:

- The right to prepare noninfringing derivative works (patched or personalized software, for instance, or remixes of media elements such as video game imagery);⁴
- The right to reverse engineer (to understand non-copyrightable elements or to create interoperable software and hardware);⁵
- Security or other research involving the software;⁶ and

¹ Consumer Review Freedom Act, S.B. 2044 (2015), *available at* <https://www.congress.gov/bill/114th-congress/senate-bill/2044/text> (last visited Feb. 12, 2016).

² Consumer Review Freedom Act § (2)(b)(1).

³ FTC Complaint for Permanent Injunction and Other Equitable Relief, 2015 WL 5754588 (M.D.Fla.) (Sept. 24, 2015), *available at* <https://www.ftc.gov/system/files/documents/cases/150928rocalabscmpt.pdf> (last visited Feb. 12, 2016).

⁴ *E.g.* Fitbit Terms of Service, *supra*.

⁵ *E.g.* Michael Aoun, *Nintendo Will Lock Your Wii if You Don't Agree to the New License Agreement* (Oct. 18, 2014), <http://www.gamnesia.com/news/nintendo-will-lock-your-wii-u-if-you-dont-agree-to-the-new-license-agreemen> (last visited Feb. 12, 2016).

⁶ *E.g.* Mary Ann Davidson, *No, You Really Can't* (Aug. 10, 2015) *available at* https://web.archive.org/web/20150811052336/https://blogs.oracle.com/maryannndavidson/entry/no_you_really_can_t (last visited Feb. 12, 2016).

- Otherwise lawful acts of circumvention⁷

The second strategy is to impose conditions on the software license. Using this approach, a vendor can write whatever they desire into a EULA and seek to enforce their dictates using copyright law. Some prohibitions imposed by vendor fiat include:

- No use of “unauthorized” hardware or software in conjunction with the device⁸ (such as third-party hardware competing with the vendor’s own peripherals, or privacy-protecting software that is common on Android systems⁹);
- No use of the licensed software on a second-hand device¹⁰ (barring resale by refusing to license the software needed for a used device to operate); and
- No continued use of the device if a new version of the EULA is available, unless it is agreed to. In other words, if the purchaser wants their property to continue functioning, they are obligated to “agree” to new terms.¹¹

Under both approaches, the vendor not only imposes unilateral contract terms on users, but seeks to turn violations of that contract into violations of copyright law, with all the baggage that comes with copyright infringement, such as statutory damages and extrajudicial takedowns.

B. EULAs and Terms of Service are Typically Contracts of Adhesion

The EULAs accompanying software-enabled devices are typically contracts of adhesion, offered on a take-it-or-leave-it basis by service providers with far greater bargaining power, legal sophistication, and time to consider the terms than their users.

This scenario is ripe for abuse, and indeed a recent study shows that the majority of Americans feel that their rights are abused by software-enabled services but feel powerless to stop it.¹² One Guardian reporter tried reading the terms of use he encountered over the course of a week and

⁷ E.g. Sony PlayStation 4 System Software License Agreement, http://doc.dl.playstation.net/doc/ps4-eula/ps4_eula_en.html (last visited Feb. 12, 2016).

⁸ Kyree Leary, *Microsoft's EULA Allows Them to Disable Pirated Games* (Aug. 2015), <http://gamerant.com/microsoft-pirate-game-disable-110/> (last visited Feb. 12, 2016).

⁹ XDA Developers, *Protecting Your Privacy: App Ops, Privacy Guard, and XPrivacy*, available at <http://www.xda-developers.com/protecting-your-privacy-app-ops-privacy-guard-and-xprivacy/>.

¹⁰ Cisco End User License Agreement, http://www.cisco.com/c/en/us/td/docs/general/warranty/English/EU1KEN_.html (last visited Feb. 12, 2016).

¹¹ *Nintendo Will Lock Your Wii*, *supra*.

¹² Joseph Turow, Michael Hennessey, and Nora Draper, *The Tradeoff Fallacy: How Marketers are Misrepresenting American Consumers and Opening Them Up to Exploitation*, Annenberg School for Communication, U. Penn. 3 (2015), https://www.asc.upenn.edu/sites/default/files/TradeoffFallacy_1.pdf.

concluded: “reading the terms and conditions simply doesn’t help ... With no negotiating power, it ends up being mostly depressing reading.”¹³ The vast majority of Americans also overestimate the degree to which statutory protections shield them from such abuse.¹⁴

It would not only be fruitless, but also *impossible* for the typical purchaser to read all of the contracts of adhesion attached to modern products and services. Each product comes with thousands of words of legal text that is supposedly binding on the purchaser. For example, the Nest EULA and Terms of Service come to nearly 11,000 words¹⁵ (not including the separate warranty document), the FitBit terms and incorporated documents are over 8,000 words,¹⁶ and the Apple Watch terms are over 5,000.¹⁷ Such figures are not uncommon.

The burden of reading these terms adds to the already overwhelming volume faced by the typical Internet user. A person would have to spend approximately 250 hours each year to read every privacy policy they encounter online.¹⁸ This amount exceeds six full work weeks and is more than the entire amount of time a typical user spends surfing the web.¹⁹ Likewise, it would be impossible for a user to read or even skim all of the terms of service they encounter.²⁰ The cumulative burden of reading these online terms and the terms accompanying software-enabled devices would be crippling to the economy – if customers actually took the time to read them.²¹

When companies choose to present a wall of legalese to their users (or put that legalese online) they know it will go unread. One company even included a cash prize for the first person to read the terms associated with their software, and it took five months and over three thousand users

¹³ Alex Hern, *I Read All the Small Print on the Internet and It Made Me Want to Die*, <http://www.theguardian.com/technology/2015/jun/15/i-read-all-the-small-print-on-the-internet> (last visited Feb. 12, 2016).

¹⁴ *Id.*

¹⁵ Nest Terms of Service, <https://nest.com/legal/terms-of-service/> (last visited Feb. 12, 2016).

¹⁶ Fitbit Terms of Service, <https://www.fitbit.com/legal/terms-of-service> (last visited Feb. 12, 2016).

¹⁷ Apple WatchOS Software License Agreement, *available at* <https://ssl.apple.com/legal/sla/docs/watchOS2.pdf> (last visited Feb. 12, 2016).

¹⁸ Aleecia M. McDonald and Lorrie Faith Cranor, *The Cost of Reading Privacy Policies*, 4:3 I/S: A Journal of Law and Policy for the Information Society 540, 560 (2008), *available at* moritzlaw.osu.edu/students/groups/is/files/2012/02/Cranor_Formatted_Final.pdf (last visited Feb. 12, 2016).

¹⁹ *Id.*

²⁰ Privacy policies are distinct from terms of service, but are a reasonable proxy because they are typically accompanied by terms of service that are just as long, or longer, and which often incorporate the privacy policy by reference.

²¹ *The Cost of Reading Privacy Policies*, *supra*, at 562.

before anyone found that line in the terms and collected the prize.²² In other words, companies impose terms unilaterally on unwitting customers, knowing that users typically are not aware of the contents of those terms.

Market forces cannot remedy these abuses for several reasons. First, the inquiry a purchaser would need to make to decide between different products' terms is prohibitive – even without hiring a lawyer. Second, the abusive terms documented below are widespread and many products are unique, meaning that alternatives are not available. Finally, because purchasers do not read EULAs and terms of service, they are not aware of particular abusive terms that might prompt them to seek alternatives.

C. Copyright Law Should Protect the Rights of Device Owners

Four principles would go a long way toward ameliorating the concerns set forth above:

- private parties cannot bootstrap ToS/EULA violations into copyright infringement;
- user rights under copyright law cannot be waived by adhesive terms;
- contract restrictions that inhibit fundamental rights should be explicitly identified, prior to purchase, in simple language;²³ and
- use of software is not an exclusive right reserved to copyright owners.

1. Private Parties Should Not Be Permitted to Bootstrap ToS/EULA Violations into Copyright Infringement

Even if EULAs and Terms of Service are considered enforceable as a matter of contract law, that would not justify escalating violations of those terms to the level of copyright infringement. A party who agrees not to reverse engineer, and does so anyway, may be liable for breach of contract, but should not be subjected to the punitive sledgehammer of copyright's statutory damages, nor should their noninfringing derivative works be subject to the extrajudicial takedown mechanisms of DMCA section 512, nor should other parties be subjected to secondary copyright liability. Even if a customer's "agreement" to EULA and ToS terms merited enforcement under contract law, transforming a breach into a violation of copyright law upsets the balance set by defenses to copyright infringement and the limited scope of the exclusive rights granted to authors.

²² The Pit Crew, *It Pays to Read License Agreements (7 Years Later)*, <http://techtalk.pcpitstop.com/2012/06/12/it-pays-to-read-license-agreements-7-years-later/> (last visited Feb. 12, 2016).

²³ Similar transparency requirements should apply to any technical restrictions attached to the software. EFF will discuss such restrictions in greater detail in its comments on 17 U.S.C. § 1201.

The Ninth Circuit has begun to address this problem by holding that contractual covenants having no nexus to copyright law cannot be converted into copyright violations by characterizing them as limits on a license.²⁴ The court observed that allowing a software copyright holder to “designate any disfavored conduct during software use as copyright infringement Would allow software copyright owners far greater rights than Congress has generally conferred on copyright owners.”²⁵ Just as a copyright term having nothing to do with copyright law should not create infringement liability (the facts before the Ninth Circuit in *MDY*), neither should contract terms that purport to erase rights explicitly preserved by copyright law (such as fair use rights to reverse engineer and create certain derivative works).

2. *User Rights Under Copyright Law Should Not Be Waived by Adhesive Terms or Magic Words*

Copyright law contains a number of protections for the owner of a copy of a work, which protect freedom of speech, personal autonomy, competition, and innovation. Some courts have made clear that transaction resembling a sale cannot be transformed into a mere license via the recitation of magic words. The Second Circuit explained in *Krause* that, regardless of the formalities of transfer, a purchaser of a copy of a copyrighted work can exercise the rights of a copy owner as long as their possession involves “sufficient incidents of ownership” over the copy, such as payment of consideration, the right to use or discard the copy, and ownership of the tangible property on which the work resides.²⁶ Unfortunately, the Ninth Circuit diverged from these decisions (including its own non-software copyright jurisprudence)²⁷ in *Vernor*, holding that calling a software transaction a license and imposing significant use and resale restrictions can strip a purchaser of resale rights.²⁸

The split between *Krause* and *Vernor* could lead to very different results in the context of software-enabled devices. Purchasers of software-enabled devices pay for tangible property that they can dispose of as they see fit, on which software copies reside. Under *Krause*, the analysis will almost certainly arrive at the correct result: owners of software-enabled devices own their copies of the software on the devices, at least for purposes of copyright law. But under *Vernor*, there is a significant risk that clever drafting of license “agreements” could undermine these important rights. The *Vernor* approach should be repudiated at the national level.

²⁴ *MDY Indus. LLC, v. Blizzard Enter., Inc.*, 629 F.3d 928, 941 (9th Cir. 2010).

²⁵ *Id.*

²⁶ *Krause v. Titleserv*, 402 F.3d 119, 124 (2d. Cir. 2005).

²⁷ *See UMG Recordings, Inc. v. Augusto*, 628 F.3d 1175 (9th Cir. 2011).

²⁸ *Vernor v. Autodesk*, 621 F.3d 1102, 1111 (9th Cir. 2010).

3. *Contract Restrictions Should be Transparent to Users, Prior to Purchase*

Contracts governing the sale of devices with embedded software that include clauses contrary to rights protected under copyright law, such as fair use, should include clear warnings to the purchaser, prior to purchase, on the product's packaging or marketing materials. Such warnings should explicitly identify, in simple language, the effect of the waiver. For example, if the EULA or TOS attached to the software in the device would inhibit a purchaser's ability to repair, re-sell, or modify her device, the label should say so clearly.

4. *Use of Software is Not an Exclusive Right Reserved to Copyright Owners*

A person interacting with a copyrighted work only needs a license to engage in acts that would, if unauthorized, violate the exclusive rights secured by copyright law. Some rightsholders argue that using software always infringes the reproduction right because portions of the copyrighted software are duplicated in a user's RAM in ordinary execution of the code. This dubious proposition has been widely criticized as a matter of statutory interpretation (it reads out the requirement that a copy exist for more than a "transitory duration") and good policy.²⁹ Copyright law should be clear that one who purchases a copy of software is entitled to run that software without needing the permission of the copyright holder, even if doing so involves making copies in the RAM of the computer or device on which the software resides. And when use of a device or software does not require a license, it would be deceptive for vendors to assert that its use is restricted based on the terms in a EULA. Allowing a manufacturer to effectively and indefinitely control the use of a device by controlling the software in that device goes beyond the traditional rights reserved to copyright owners and upsets the balance between rightsholders and device owners, in part by providing a hook for restrictive EULA terms.

D. Conclusion

A uniform system of federal copyright requires reining in the mischief of EULAs and other contracts of adhesion. The suggestions above represent the first steps toward restoring a rational balance between the rights granted to copyright owners and the rights of those who own copies of such works.

²⁹ E.g., Aaron K. Perzanowski. *Fixing RAM Copies*, 104 Northwestern Univ. L. Rev. 1066. n.85 (2010), available at http://scholarlycommons.law.case.edu/cgi/viewcontent.cgi?article=1045&context=faculty_publications (last visited Feb. 12, 2016).

III. API Copyrightability Would Threaten Competition and Innovation for Software-Enabled Devices

An Application Programming Interface (API) is a standardized specification for software routines, telling programmers what functional behavior their software must have to successfully communicate according to the API protocol. In essence, it is a language for different software modules to communicate, enabling programmers to make interoperable systems without needing to create a custom interface at each interconnection. As long as two systems implement the functional specification that is the API, they can successfully interoperate.

Despite the fact that an API specification is simply a functional system of intercommunication, the Federal Circuit recently opened the door for API copyrightability, overruling a sensible district court decision that held uncopyrightable the “utilitarian and functional set of symbols” that constitute an API.³⁰ The panel’s decision threatens to choke off a longstanding and critical tradition of innovation that relies on making devices and software interoperable by reimplementing the functional properties that constitute an API.

The freedom to reimplement and extend existing APIs has been key to the development of software-enabled devices. It made possible the emergence and success of modems, printers, microcontrollers, wireless communication between devices, the Internet of Things, and general-purpose computers—by ensuring that competitors could challenge established players and advance the state of the art.

The widespread availability of diverse, cheap, and customizable computers owes its existence to the lack of copyright on the specification for IBM’s Basic Input/Output System (BIOS).³¹ Companies such as Compaq and Phoenix reimplemented IBM’s BIOS without fear of copyright claims, making possible PC clones and improvements in portability and function.³² Open APIs were also essential to many modern computing developments, including those of operating systems such as UNIX,³³ programming languages such as “C”,³⁴ the Internet’s network

³⁰ *Oracle America, Inc. v. Google Inc.*, 750 F.3d 1339 (Fed. Cir. 2014).

³¹ Charles H. Ferguson & Charles R. Morris, *Computer Wars: The Fall of IBM and the Future of Global Technology* 27–29 (1994).

³² *Id.* at 53–55; see Van Lindberg, *Intellectual Property and Open Source: A Practical Guide to Protecting Code* 240–41 (2008).

³³ Heather J. Meeker, *The Open Source Alternative* at 3–4 (2008).

³⁴ For the C API specification and some of its many implementations, see P.J. Plauger, *The Standard C Library* 3 (1991); C Run-Time Libraries, Microsoft Developer Network, *available at* [http://msdn.microsoft.com/en-us/library/abx4dbyh\(v=vs.80\).aspx](http://msdn.microsoft.com/en-us/library/abx4dbyh(v=vs.80).aspx); *The Native Android API*, Mobile Pearls, *available at* <http://mobilepearls.com/labs/native-android-api/>; *The GNU C Library* (glibc), The GNU Project, *available at* <http://www.gnu.org/software/libc/>.

protocols,³⁵ and cloud computing.³⁶

Uncopyrightable APIs spur the creation of software that otherwise would not have been written. When programmers can freely reimplement or reverse engineer an API without the need to negotiate a costly license or risk a lawsuit, they can create compatible software and devices that the interface's original creator might never have envisioned or had the resources to create. Moreover, compatible APIs enable people to switch platforms and services freely, and to find devices that meet their needs rather than being locked in to a device "ecosystem."³⁷ Without the compatibility enabled by API compatibility, device owners could also be forced to leave their data behind when they switch to a new device.³⁸

The freedom to reimplement APIs also helps rescue "orphan" devices—systems whose creators have either gone out of business or abandoned their product in the marketplace.³⁹ Government entities and non-profits are especially susceptible to the orphan programs problem as they often cannot afford to upgrade and are left using legacy technologies for years or decades. When a popular computer platform or service shuts down, the ability to freely reimplement APIs protects the communities that rely on that software.

Copyright law should continue to exclude APIs from the scope of copyrightable subject matter and preserve the settled business practices that have enabled software-enabled devices to flourish.

³⁵ Kaare Christian & Susan Richter, *The UNIX Operating System* 6, 502 (3d ed. 1994); *Programming with TCP/IP—Best Practices*, 3 HP OpenVMS Technical Journal 3, 5 (2004).

³⁶ *About AWS*, Amazon.com, <http://aws.amazon.com/what-is-aws>.

³⁷ For example, Wine is a compatibility layer that reimplements the Windows API so that third-party Windows programs can run on UNIX-based operating systems like Linux and Mac OS X. *About Wine*, WineHQ, available at <http://www.winehq.org/about/>. Microsoft (which owns the copyright in the Windows operating systems) is hostile to Wine and API copyrightability could provide a hook for it to prevent its use or development, making a host of software less useful by restricting it to computers running the Windows operating system. *Microsoft Admits Targeting Wine Users*, ZDnet (Feb. 25, 2005), available at <http://www.zdnet.com/microsoft-admits-targeting-wine-users-3039189180/>.

³⁸ Alan Henry, *How to Upgrade to a New Android Phone and Take Everything with You*, available at <http://lifehacker.com/5843206/how-to-upgrade-to-a-new-android-phone-and-take-everything-with-you>.

³⁹ See Bobbie Johnson, *Oh, Delicious—Where Did It All Go So Wrong?*, GigaOm (Sept. 28, 2011), available at <http://gigaom.com/2011/09/28/oh-delicious-where-did-it-all-go-so-wrong>; *A Tour of Pinboard*, Pinboard available at <http://pinboard.in/tour#api>.